Supplemental Appendix

The Inequality Multiplier: Market Inelasticity and the Persistence of Wealth Inequality

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1. Inelastic Investors

As shown in Figure 1A, the share of wealth held by the top 1% in the United States has risen from around 23% in the mid-1970s to 35% in 2021 (Saez and Zucman (2016)). Over the same period, the fraction of the total U.S. equity market held by institutional investment funds (such as Mutual Funds, ETFs, and Insurance funds) has risen from around 25% to roughly 40% (peaking near 47% around 2000). These investors are characterized by a relatively low price elasticity of demand for equity (or inelasticity), which leads to a relatively more stable equity share compared to their elastic counterparts (such as hedge funds and households) over the business cycle (Figure 1B). The growing presence of inelastic investors makes the aggregate demand curve in equity markets more inelastic over time, implying that changes in household or institutional equity demand move asset prices.

To make this point more clearly, Figure 2 plots the cross-correlation between the change in top 10% (or top 1%) wealth share against a measure of the change in the macro-market elasticity[1] over time in the U.S. Each point represents a quarter between 1990 and 2021. Both plots demonstrate that over the past three decades, as market elasticity has reduced, wealth inequality has increased. The purpose of this paper is to understand to what extent inelastic markets can explain the increase in wealth inequality.

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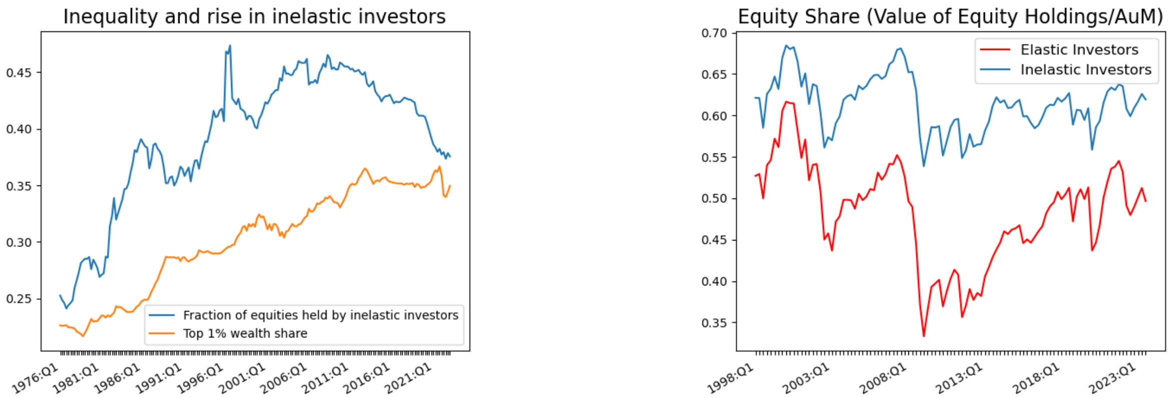
[1]The measure of elasticity is derived from Gabaix and Koijen (2021) and constructed by dividing the value of debt holdings (B) of inelastic investors by the value of their equity holdings (PQ); as their mandate tilts further toward equity, B/PQ falls and the market becomes more inelastic. We motivate this measure of elasticity further in our model.



A. Rise in inequality and inelastic investors **B.** Portfolio behaviour of investors

FIGURE 1. Panel A plots the top 1% wealth share (rising from around 23% to 35%) together with the fraction of U.S. equities held by inelastic investors (rising from around 25% to roughly 40%). Panel B shows that inelastic investors hold a higher and more stable equity portfolio share—equity holdings over assets under management—than their elastic counterparts.

Source: Authors' calculation from Flow of Funds Data and Saez and Zucman (2016). Inelastic Investors include Mutual Funds, Closed-end funds, ETFs, Government Retirement Funds, Insurance funds.

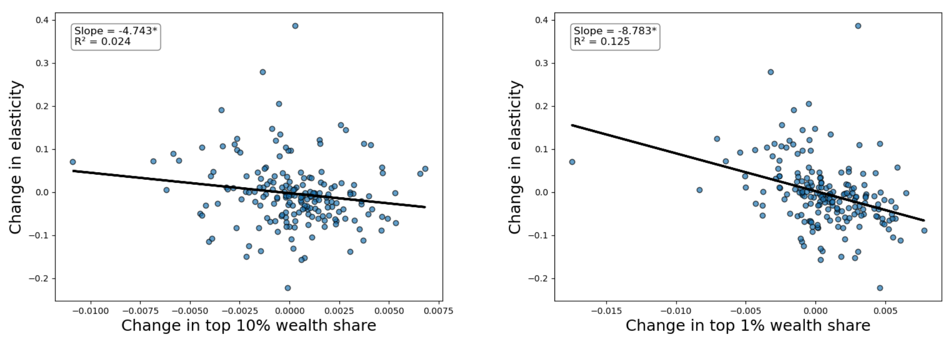


FIGURE 2. Panel A shows a significant negative correlation between the change in top 10% wealth share between 1990 and 2021 and change in market elasticity. Panel B shows a larger negative correlation between the change in top 1% wealth share and change in market elasticity.

Note: Inelastic Investors include Mutual Funds, Closed-end funds, ETFs, Government Retirement Funds, Insurance funds. In Appendix 1, we plot the actual time series for the share of equities held by inelastic investors, and how inelastic investors are characterised by more stable equity portfolio shares relative to their elastic counterparts.

Source: Quarterly top 10% and top 1% wealth shares from Saez and Zucman (2016) and B/PQ from flow of funds data (between 1990 and 2021).

2. Micro-foundations for Market Inelasticity

The inelasticity of equity markets, which we take as given, admits several complementary microfoundations—intermediary mandates and agency frictions, transaction costs and slow-moving capital, preferred-habitat demand, and binding intermediary constraints—which we review in turn.

Institutional constraints and mandates contribute significantly, as shown by Chinco and Sammon (2024). For example, mutual funds, which own 22% of U.S. equities, and ETFs, which own 6.4%, are routinely subject to explicit or implicit investment mandates which reduce their ability to exercise discretion in day-to-day portfolio allocation. He and Xiong (2013) use an agency-based model to argue that tight investment mandates are justified for most fund managers except those with exceptional talents, using this to argue for why we empirically observe the use of stringent investment mandates.

Other micro-foundations include the transactional costs of moving wealth between liquid and illiquid assets in Kaplan et al. (2018), which reduces the ability of households to seamlessly adjust portfolio shares invested in different asset classes. Slow-moving capital in Duffie (2010) (due to institutional impediments to trade such as search costs, time to raise capital or investor inattention) can generate sharp price reactions in the short- to medium-run. Preferred habitat investors in Vayanos and Vila (2021) and Chen et al. (2012) create limits to arbitrage, thereby implying that asset markets are not fully elastic. More generally, behavioral frictions such as inattention may also help explain limited ability of investors to absorb large flows into and out of asset markets.

Haddad and Muir (2021) use time-varying risk aversion of financial intermediaries along with costly direct investment in asset markets by households to show that the elasticity of household asset demand to asset prices is decreasing in the risk aversion of financial intermediaries as well as the cost of direct investment, generating inelastic demand curves. In a similar vein, He and Krishnamurthy (2018) review approaches under which institutional frictions such as equity capital constraints (He and Krishnamurthy (2013)), endogenous borrowing constraints (Brunnermeier and Sannikov (2014)), value-at-risk regulations (Adrian and Shin (2010)) and leverage regulations (Du et al. (2023)) create downward sloping demand curves for assets, specifically for large and well-regulated financial intermediaries such as commercial banks, investment banks and hedge funds.

These results carry over to other financial institutions such as mutual funds and asset managers, focusing in turn on skills of mutual fund managers (Kacperczyk et al. (2016)), limits to information acquisition (Gârleanu and Pedersen (2018)) and benchmarking considerations (Basak and Pavlova

(2013), Pavlova and Sikorskaya (2023)), all of which result in the reduced ability (or appetite) of asset managers to adjust portfolios frequently and rapidly, thereby generating inelasticity in asset markets.

A natural question is why macro-arbitrageurs have not emerged to take advantage of all the above frictions, thereby offsetting the inelasticity generated due to these frictions. Gabaix and Koijen (2021) document limited size and influence of arbitrageurs, showing that they account for less than 5% of the overall equity market holdings. They also note that hedge funds seem to respond to crisis episodes by selling assets rather than buying, the transfer of equity risk across different types of investors is minimal, and households' equity share has shown little variation over time. Finally, Haddad et al. (2025) highlight that competitive strategic response of investors to other investors reducing their elasticity of demand is limited, further contributing to market inelasticity. These suggest that forces which might offset market inelasticity are limited.

Taken together, there is ample evidence for the inelasticity of equity markets, which we take as given. For ease of modelling, we introduce inelasticity through investment mandates. However, any of the above features could be incorporated to generate qualitatively similar results.

3. Quantitative Model: Full Impulse Response Set

This section reports the full set of impulse responses from the calibrated quantitative model, complementing the headline flow-demand-shock responses in the main text (Figure 2) and the government-borrowing and middle-LTV responses collected in Appendix G. We first give the dividend shock in the same four-panel format, and then present each core variable in a grid across all shocks. Throughout, the inelastic model is shown as a solid line and the elastic model as a dashed line; responses are to a positive one-standard-deviation innovation over a sixty-period horizon.

4. Sensitivity of the Wealth-Share Transition to the Mandate

We assess how the simulated 1989–2023 wealth-share transition depends on the two parameters that govern the intermediary's mandate: the target equity share θ^* and the rigidity χ^{MF}. For each,

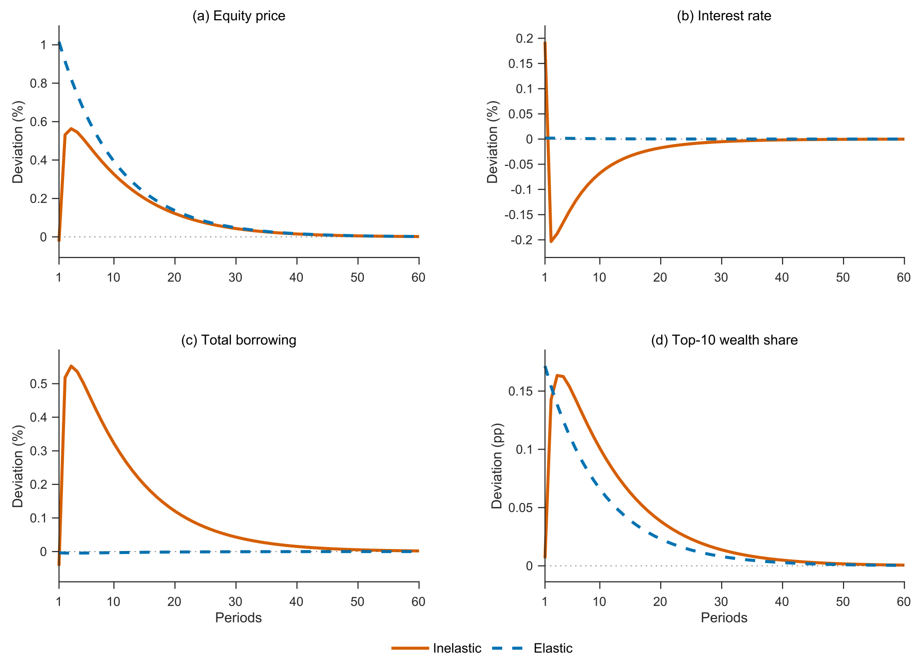


FIGURE 3. Dividend shock: headline responses

Note: Response to a positive one-standard-deviation dividend shock, inelastic (solid) versus elastic (dashed). Panel (a) equity price; (b) safe rate; (c) total household borrowing; (d) top-10 wealth share. Panels (a)–(c) are percentage deviations; panel (d) is in percentage points.

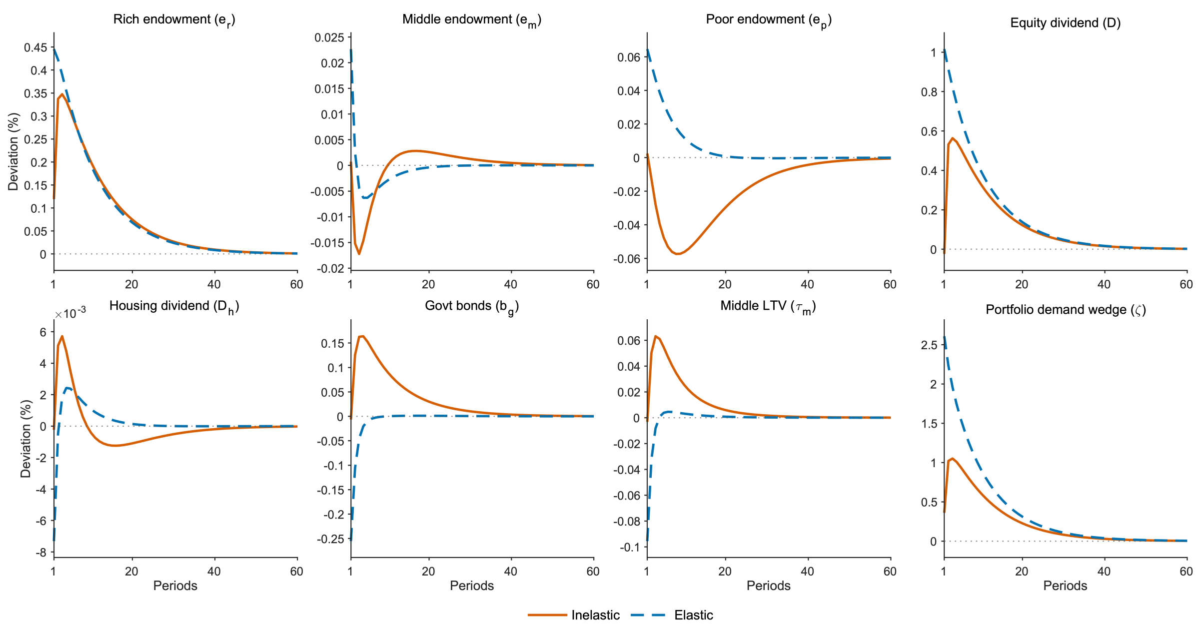


FIGURE 4. Equity price across all shocks

Note: Percentage-deviation response of the equity price to each shock, inelastic (solid) versus elastic (dashed), over sixty periods.

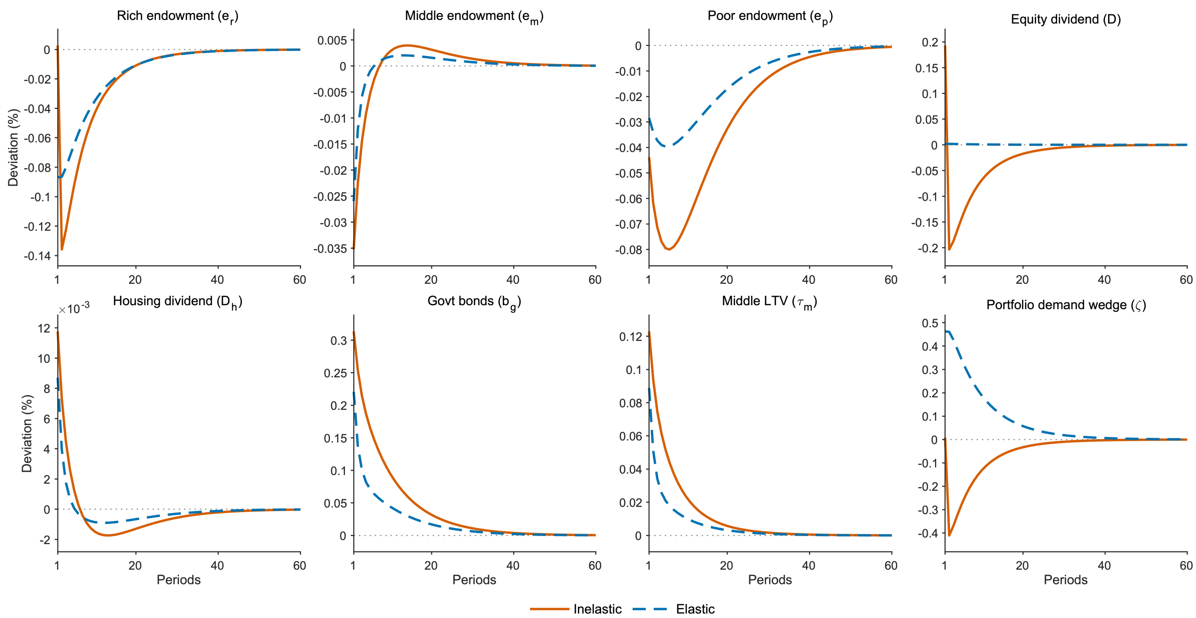


FIGURE 5. Safe interest rate across all shocks

Note: Percentage-deviation response of the safe rate to each shock, inelastic (solid) versus elastic (dashed).

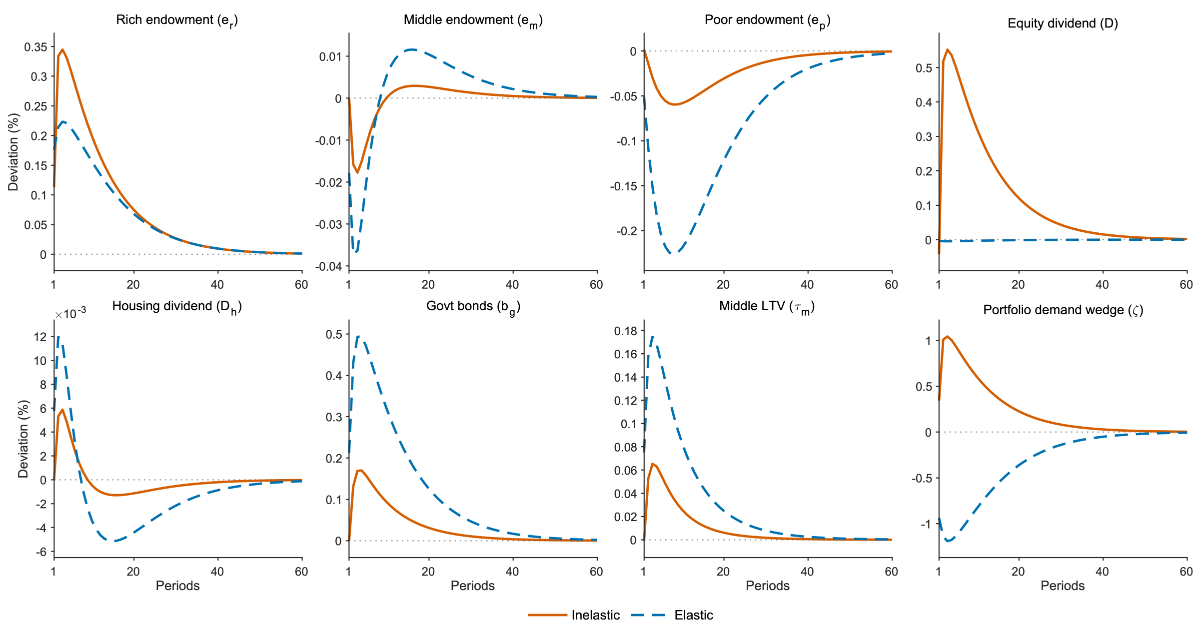


FIGURE 6. Total household borrowing across all shocks

Note: Percentage-deviation response of total household borrowing to each shock, inelastic (solid) versus elastic (dashed).

we re-solve the 1989 reference steady state at a value below and a value above the calibrated one—re-hitting the wealth-share targets—and re-run the perfect-foresight simulation, holding all other parameters fixed.

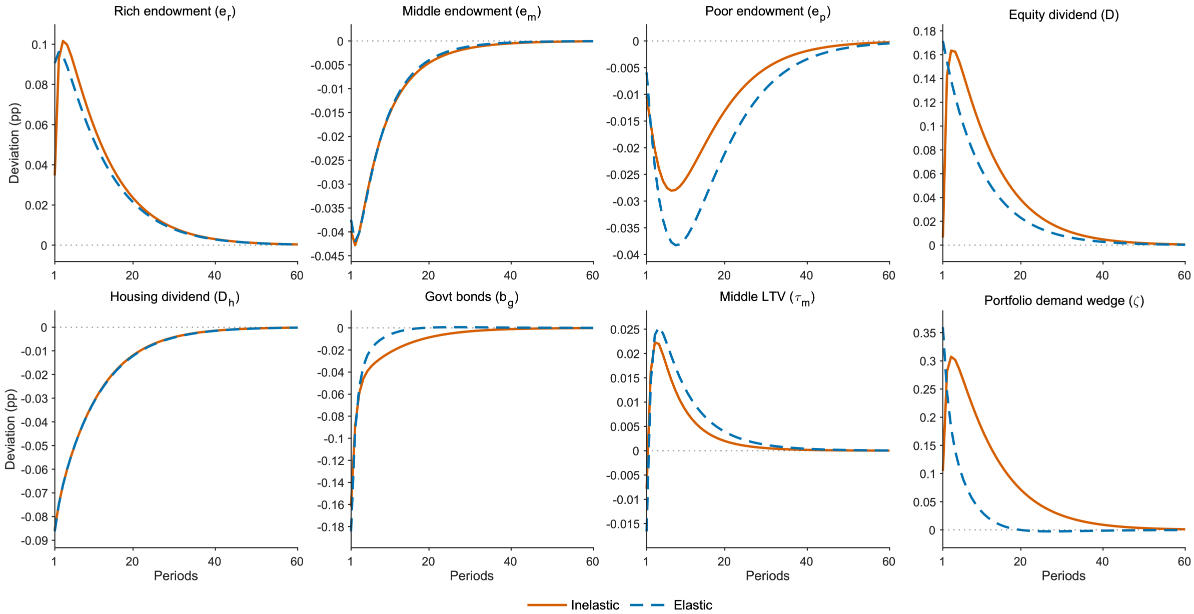


FIGURE 7. Top-10 wealth share across all shocks

Note: Response of the top-10 wealth share (percentage points) to each shock, inelastic (solid) versus elastic (dashed).

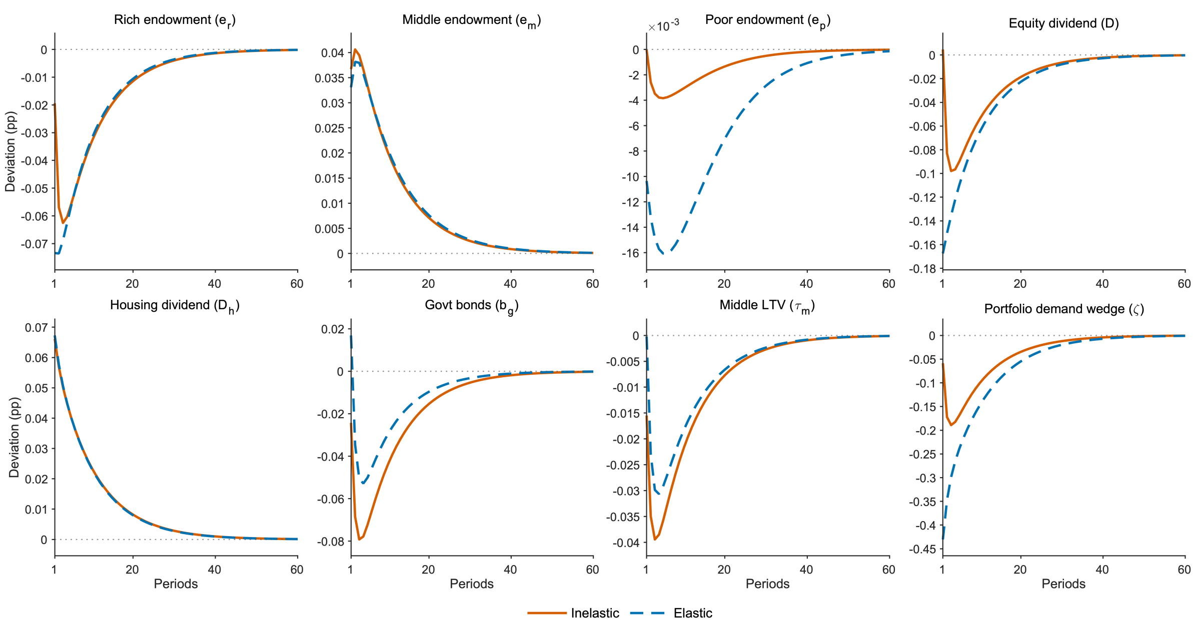


FIGURE 8. Middle-40 wealth share across all shocks

Note: Response of the middle-40 wealth share (percentage points) to each shock, inelastic (solid) versus elastic (dashed).

The transition is essentially invariant to χ^{MF} over its inelastic range: once the mandate binds, the wealth-share paths are indistinguishable across $\chi^{MF} \in \{20, 171, 500\}$. What matters for the dynamics is that the market is inelastic, not the precise degree of rigidity, so we do not plot the

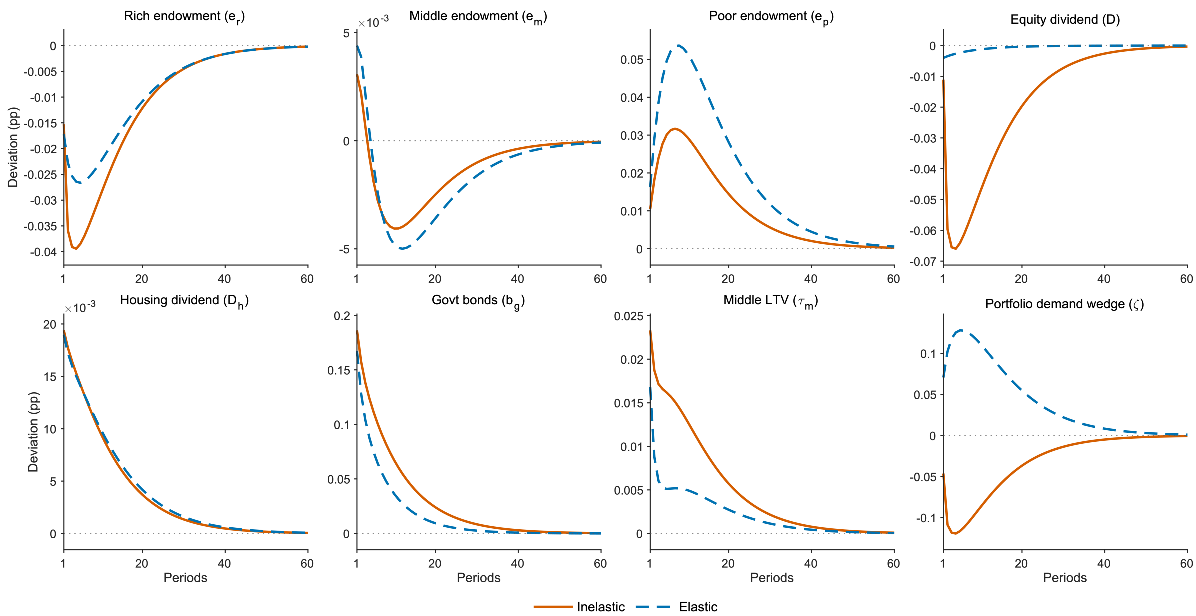


FIGURE 9. Bottom-50 wealth share across all shocks

Note: Response of the bottom-50 wealth share (percentage points) to each shock, inelastic (solid) versus elastic (dashed).

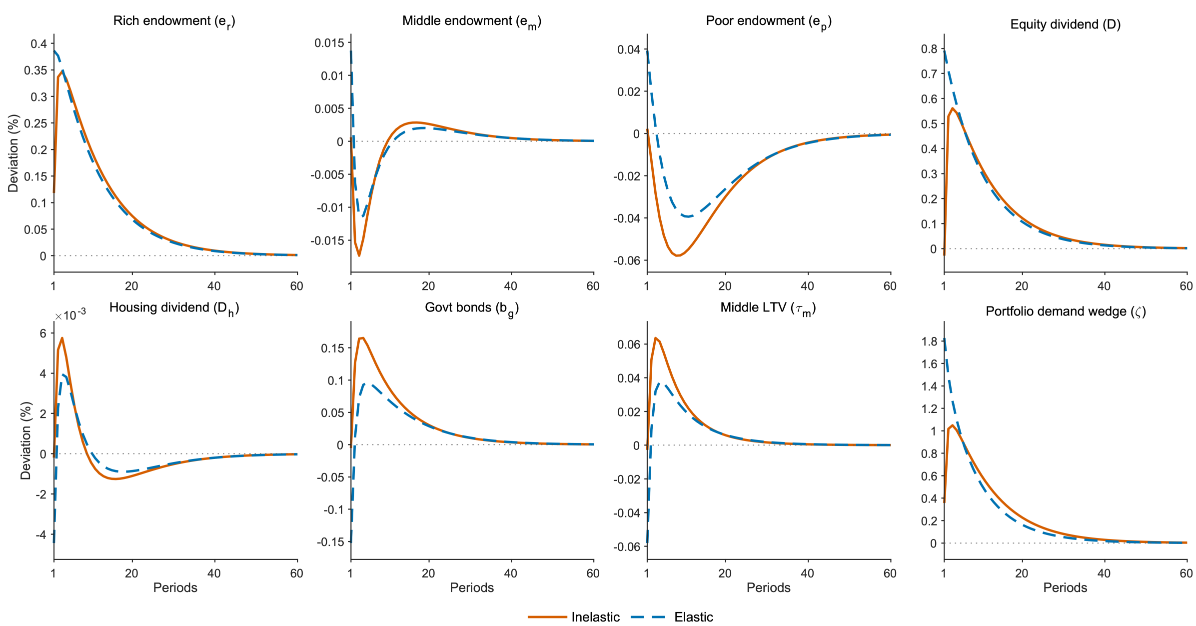


FIGURE 10. Wealthy equity holdings across all shocks

Note: Percentage-deviation response of wealthy equity holdings to each shock, inelastic (solid) versus elastic (dashed).

χ^{MF} comparison.

The target equity share θ^*, by contrast, shapes the transition through the borrowing channel. Figure 14 plots the simulated top-10%, middle-40%, and bottom-50% wealth-share paths for $\theta^* \in$

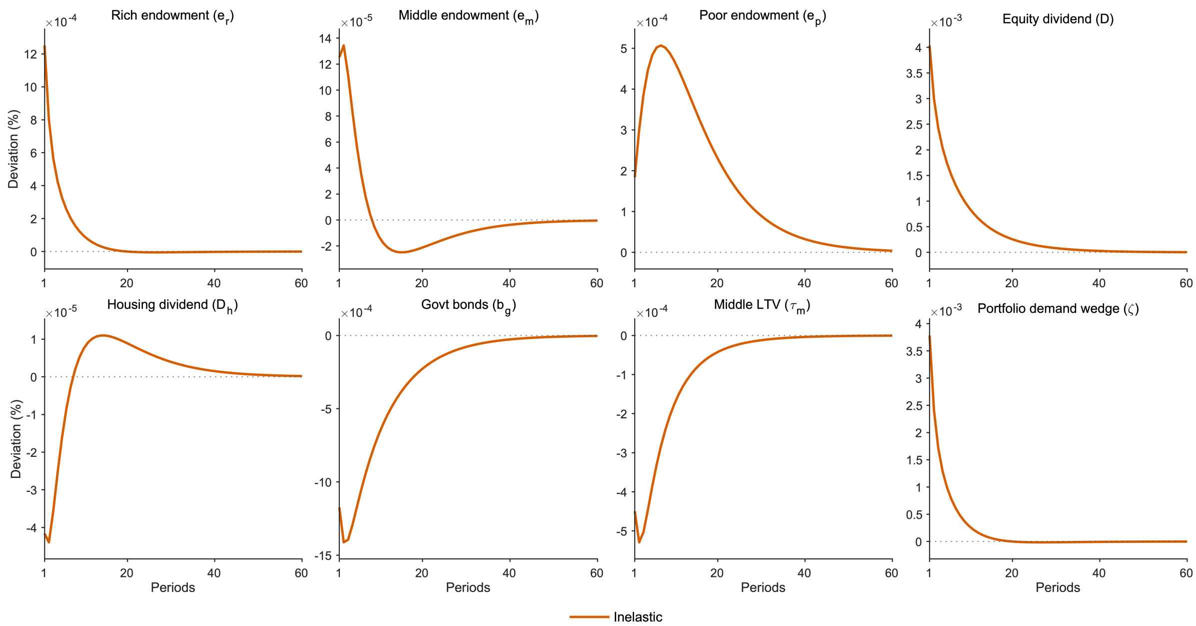


FIGURE 11. Mandate share θ across all shocks

Note: Percentage-deviation response of the mandate share θ to each shock. Only the inelastic model is shown (the elastic model has no binding mandate).

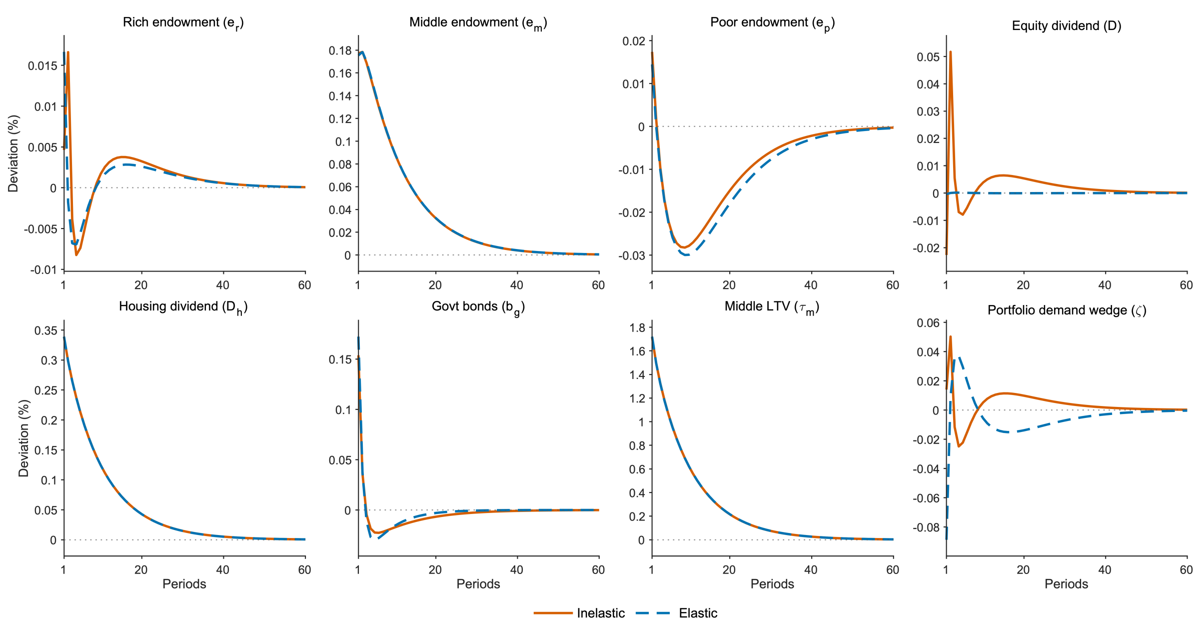


FIGURE 12. Middle-class borrowing across all shocks

Note: Percentage-deviation response of middle-class borrowing to each shock, inelastic (solid) versus elastic (dashed).

$\{0.70, 0.78, 0.86\}$ against the data. A lower θ^* leaves the fund holding a larger bond share $1 - \theta^*$, so a given set of shocks is intermediated into more credit and the borrowing channel is correspondingly stronger. Its clearest imprint is at the bottom of the distribution (panel c): at $\theta^* = 0.70$ the bottom-

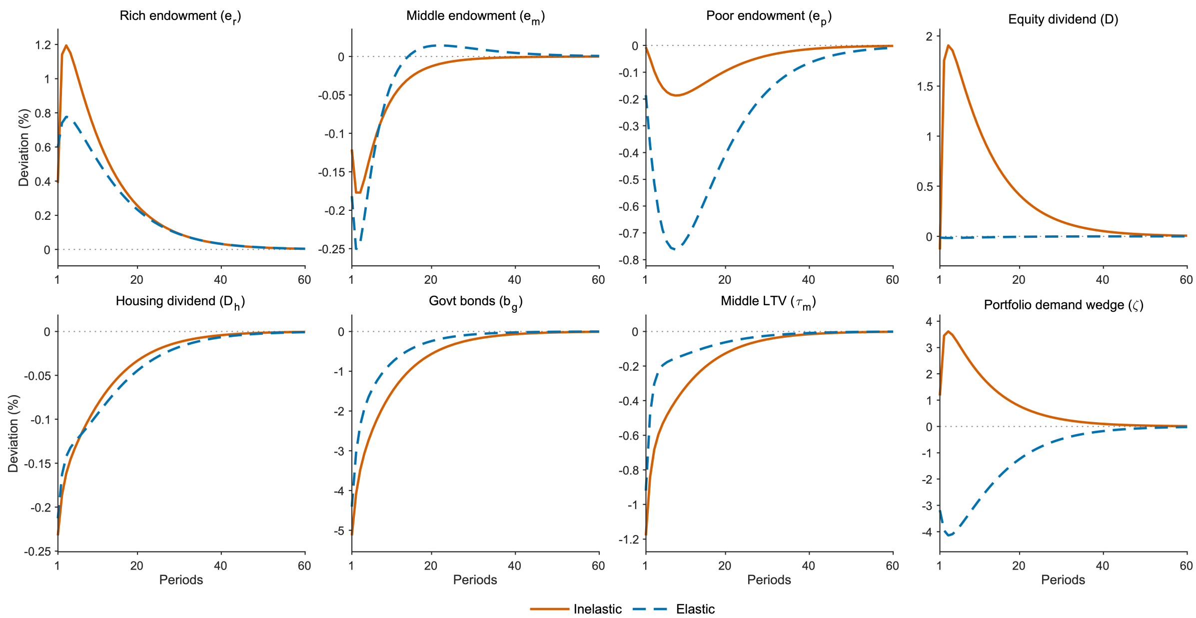


FIGURE 13. Bottom-50 borrowing across all shocks

Note: Percentage-deviation response of bottom-50 borrowing to each shock, inelastic (solid) versus elastic (dashed).

50% share falls further through the mid-2000s—a trough near −4.9 percentage points around the crisis, against −2.7 at $\theta^* = 0.86$—and rebounds more sharply after 2020. The middle-40% share (panel b) shows the same ordering, while the top-10% share (panel a), governed mainly by the equity-investment channel, is far less sensitive. The calibrated value $\theta^* = 0.78$ lies between the two.

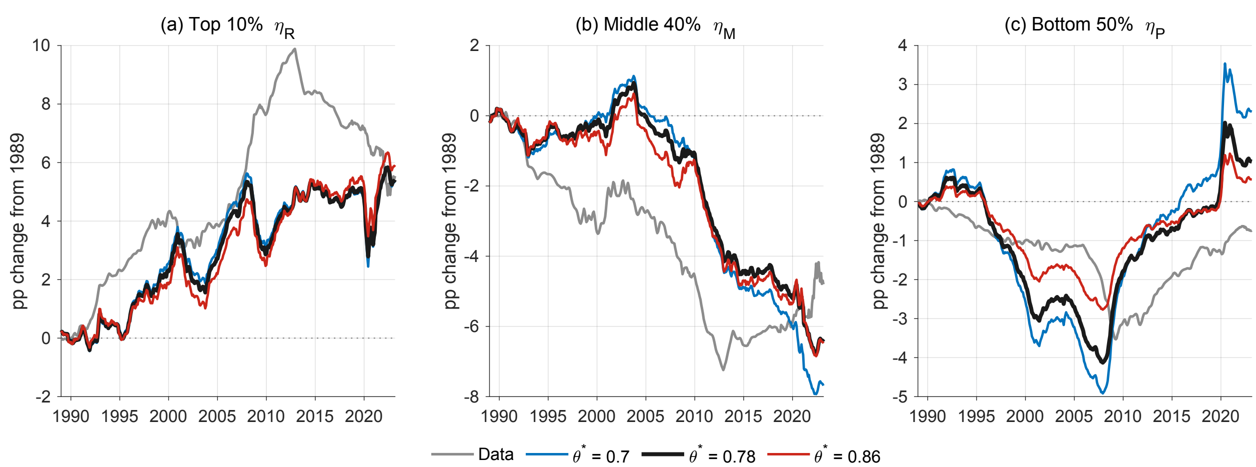


FIGURE 14. Sensitivity of the simulated wealth-share transition to the fund's target equity share θ^*.

Note: Perfect-foresight simulation of the top-10%, middle-40%, and bottom-50% wealth shares (percentage-point change from 1989) for $\theta^* = 0.70$ (below calibration), 0.78 (calibrated, bold black), and 0.86 (above), against the data (grey). Each path re-solves the 1989 reference steady state at the given θ^*, re-hitting the wealth-share targets, and re-runs the transition with all other parameters held at their calibrated values.